Learning Cognitive and Linguistic Prosodic Categories for Automatic Cross-Lingual Sign Language Understanding

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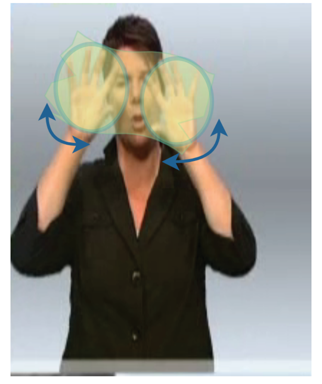
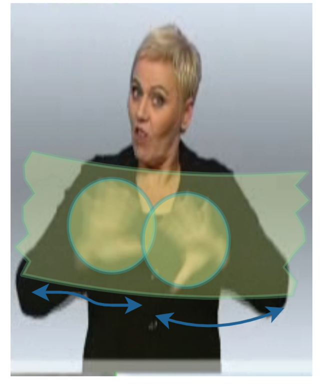
Motivation

Elements of prosody such as rhythm, stress, or lengthening play important roles in distinguishing meaning and signaling intensification in sign language, similar to spoken languages.





It is important for sign language understanding (SLU) systems to be able to learn accurately from the data and generate presentations that respect prosody.

SLU systems rely on gloss representations —a shortened approximation of spoken language that has mappings to signs. However, due to its sparse form, gloss annotations often do not capture prosody of sign language. Inan et al. (2022) show that prosodic information and intensifiers can improve sign language generation in German Sign Language (DGS). Motivated by this, we would like to analyze the understanding of intensifiers in a multilingual setting.

Our goal in this work is to improve modeling prosody in SLU systems by developing strategies grounded in the literature of linguistics to model **intensification** in gloss annotations.

Approach

Inan et al (2012) apply cognitive strategies of intensification from American Sign Language (ASL) to DGS, showing that cognitive prosodic capabilities are transferrable across languages. We explore whether a gloss intensity classifier is also transferable across language domains:

- We gather sign language video, transcription and gloss data from PHOENIX-14T and BBC-Oxford British Sign Language (BOBSL) datasets.
- For British Sign Language (BSL) dataset, we generate missing Gloss tokens using state-of-the-art gloss generation models (Kayo et al. 2020).
- We run intensifier classifier proposed by Inan et al (2022) on the PHOENIX-14T and BOBSL datasets.
- We compare classifier results for DGS and BSL for intensification
- Analyze the effects of the domain of the dataset on the intensifier abundance and the changes of intensifier use across different sign languages and draw parallels with shared cognitive skills of prosody across languages.

Learning Prosody Through Intensification

Linguistic and Cognitive Prosodic Categories

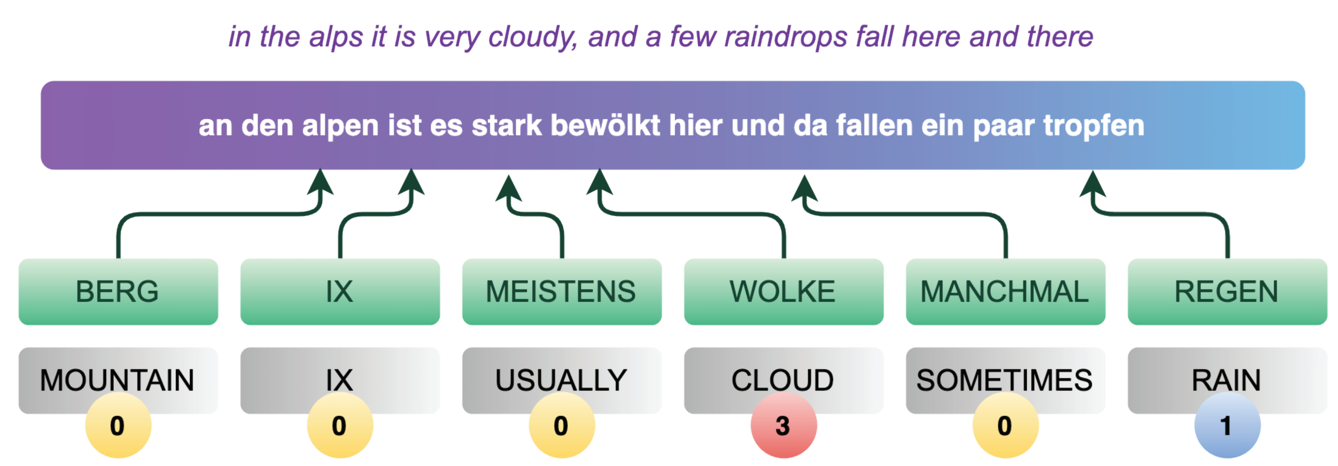
End-marking: Wilbur et al. (2012) explain that depending on the degree of the adjective, there is a "sharp movement to a stop" in the final timing of a sign.

Delayed Release: Wilbur et al. (2012) show that the initial time interval of a sign also gets modified with a slight pause in the beginning and a faster continuation of the sign.

Reiteration: Nicodemus et al. (2014) describe that during the end-marking and elongation phase, a sign might be reiterated multiple times to mark the intensification.

Suffixation: In other sign language datasets such as Public DGS Corpus uses a gloss annotation convention where the phonemes and synonyms that have different signs contain a number that is added as a suffix to the end of the gloss (Konrad et al., 2020).

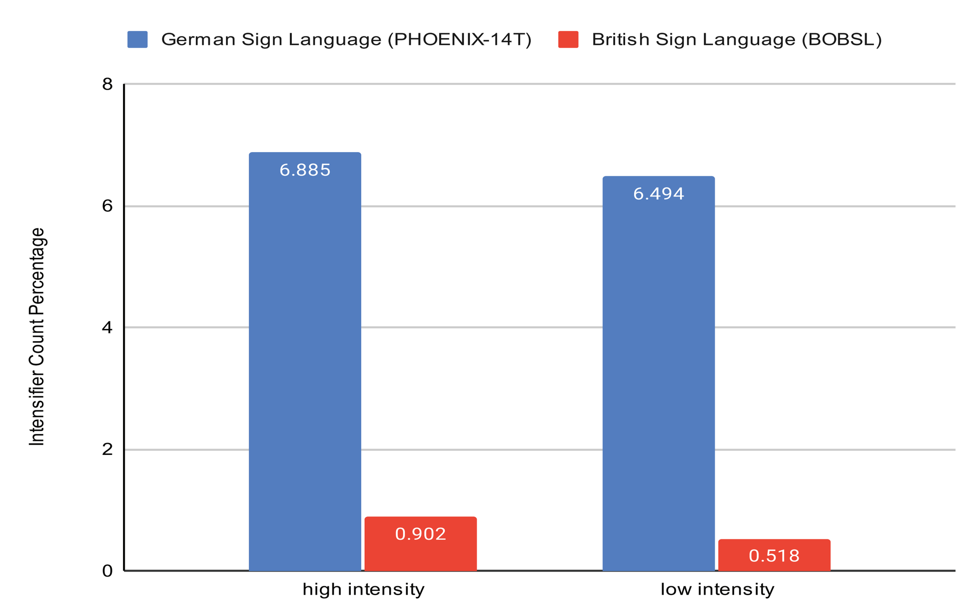
Identifying Prosodic Categories in Sign Language Data



We follow a similar strategy to Inan et al (2022) while classifying the glosses into three different intensity levels. In order to do so, we first generate gloss if it is missing in the data (especially for BSL), then separate each gloss token in a single utterance, and then we use a multilingual BERT model to extract features from the text and then classify them into 0, 1 or 2, for no intensity, low intensity and high intensity classes respectively.

Multi-Lingual Data Exploration

Prosody Changes from German to British Sign Language



	Adjective Percentages	
	PHOENIX-14T	***BOBSL***
Transcript	0.94	1.83
Gloss	0.35	1.40

There is a difference between BSL and DGS intensifier and adjective count percentages in the generated glosses and classifications. This is due to bot systematic errors and differences between languages.

Qualitative Analysis

We manually analyzed 200 pairs of texts, their corresponding GLOSS translations and predictions by our cross-lingual intensification models.

Key observations:
Some errors clearly result from limitation of multilingual models.

Example: In BSL gloss token "light" is not tagged as low intensity. This can be due to the difference in context where words appear in different sign languages. In the DGS data, they appear together in the context of weather such as "light rain", but in the BSL data, they can appear in everyday conversations such as "light bulb".

Errors can result in automatic translations from text to gloss.

Example:

BSL Transcript	Translated Gloss
Yes, thanks very much	APPLAUSE AND THANKS VERY CONSIDERATE

The models can pick up intensity markers for words in the transcript that can clearly quantify object(s), e.g., "bit", "few" and "much".

Discussion & Conclusion

Even though cognitive skills are transferable across languages, this does not manifest in machine learning classifiers for intensification understanding in sign language.

This is observable in both our qualitative and quantitative results. There are differences between DGS and BSL. These are caused by the classifier not being pre-trained on BSL but DGS, which biases the output. In addition, there are language-specific biases that result in differences of intensifier and adjective counts.

In order to push the field of sign language research, an orcestrated effort from neuroscience, cognitive science, computer science, and linguistics is needed. Lack of publicly-available resources is a major concern.

Despite limitations, we show that the strategies of intensification, grounded in the linguistics and cognitive science literature of signed languages, ave the abiltiy to provide more quantitive insights into the sign language understanding.

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